
MHF 4U1

CHAPTER 3: RATIONAL FUNCTIONS

# of Days	Section	Homework	Completion
1	3.1 The Reciprocal of a Linear Function	Page 153 -155 # 1-3, 5, 7-9, 12	
2	3.2 Reciprocal of a Quadratic Function	Page 164-166 # 1, 2, 4, # 5(not v), 8, 10, 11, 14	
2	3.3 Rational Functions in the Form $f(x) = \frac{ax+b}{cx+d}$	Page 174 #1-3 eo, 5-9	
2	3.5 Making Connections Day 1: Special Cases (Holes) Day 2: Special Cases (Oblique Asymptotes)	Day 1: Page 190 #4,5abd Day 2: Page 190 #8,9	
2	3.4 Solving Rational Equations and Inequalities	Day 1 : Page 183 #(2-5)eo, 9-12, 16, 18 Day 2 : Handout	
2	Review	Page 192 #1 – 5, 7 –11, 12ab, 13, 14, 16 Handout	
1	CHAPTER 3 TEST		

CHAPTER 3 TEST DATE:

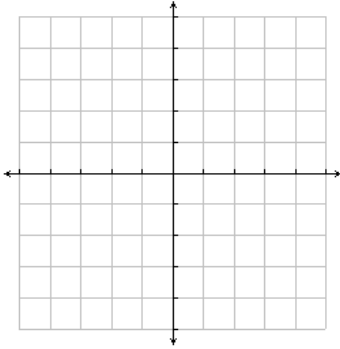
3.1 THE RECIPROCAL OF A LINEAR FUNCTION

When you divide two polynomial functions, the resulting function is a RATIONAL FUNCTION.

$$f(x) = \frac{P(x)}{Q(x)}, \text{ where } P(x) \text{ and } Q(x) \text{ are both polynomial functions and } Q(x) \neq 0.$$

Recall:

Sketch $f(x) = \frac{1}{x}$. State the D & R.

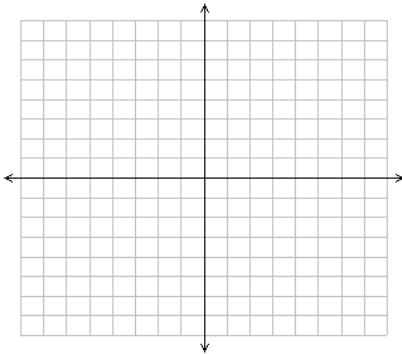


In general:

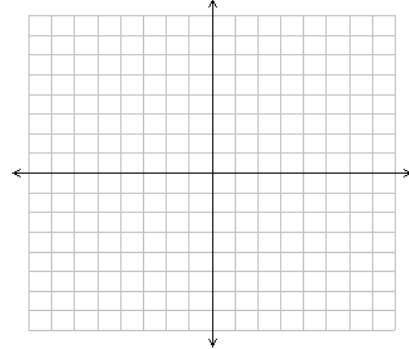
- $f(x) = \frac{1}{kx - c}$ is the reciprocal of a linear function.
- The restriction on the DOMAIN of a reciprocal linear function can be determined by finding the value of x that makes the denominator zero.
- The VERTICAL ASYMPTOTE has equation of the form $x = \frac{c}{k}$.
- The HORIZONTAL ASYMPTOTE of a reciprocal function has equation $y = 0$ (assuming there is no vertical shift).

Ex. 1: Sketch the following reciprocal functions. Label your work.

$$f(x) = \frac{1}{2x - 8}$$



$$g(x) = \frac{1}{-2x - 6}$$



Ex. 2: Using the graph of $f(x) = \frac{1}{2x - 8}$ from above, describe the behaviour of the function as x approaches each key value.

As $x \rightarrow$	$f(x) \rightarrow$
4^+	
4^-	
$+\infty$	
$-\infty$	

NOTE: (notation)

- As $x \rightarrow +\infty$ means "as x approaches positive infinity"
- As $x \rightarrow -\infty$ means "as x approaches negative infinity"
- As $x \rightarrow a^+$ means "as x approaches a from the right"
- As $x \rightarrow a^-$ means "as x approaches a from the left"

Ex. 3: Refer to the graphs of $f(x)$ and $g(x)$ from example 1.

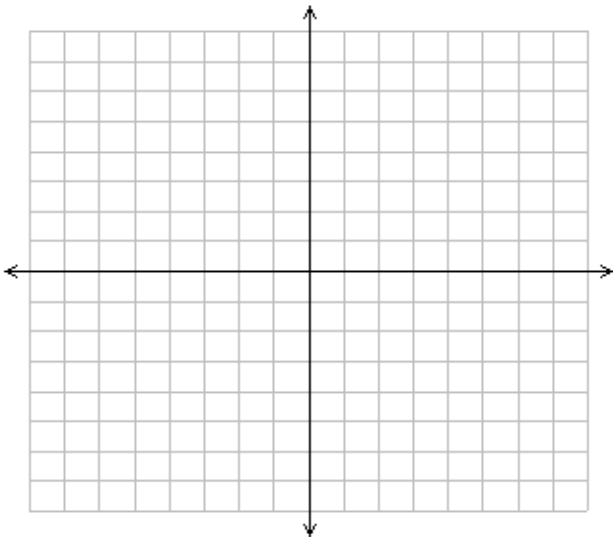
Sketch a tangent on each branch of $f(x)$. Are the slopes of the tangents positive or negative?

Sketch a tangent on each branch of $g(x)$. Are the slopes of the tangents positive or negative?

Notice that when the slope of a tangent is _____, the rational function is always _____. Also, when the slope of a tangent is _____, the rational function is always _____.

Ex 4: Given the function: $f(x) = -\frac{3}{x-2}$.

- a) Sketch a graph of the function & label the major characteristics. State the domain and range. State the x- and y-intercepts.



- b) Describe the behaviour of the function near the restriction on x using the table provided.

as $x \rightarrow$	$f(x) \rightarrow$

The function approaches a _____,
but does not _____. So, the curve is
_____ at this line. This line is called a
_____ and has equation _____.

- c) Describe the end behaviour of the function using the table provided.

as $x \rightarrow$	$f(x) \rightarrow$

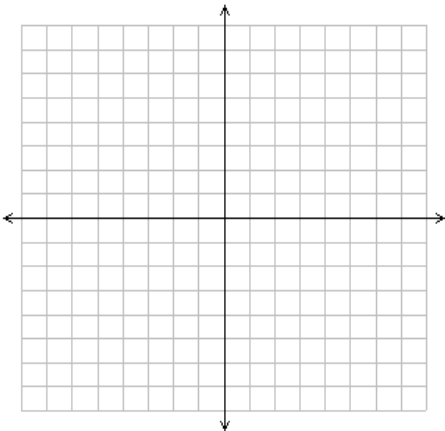
The function approaches a _____,
but does not _____. So, the curve is
_____ at this line. This line is called a
_____ and has equation _____.

Ex 5: Determine an equation of a linear reciprocal function, in the form $f(x) = \frac{1}{kx - c}$, with a vertical asymptote at -4 and y-intercept at $\frac{1}{2}$.

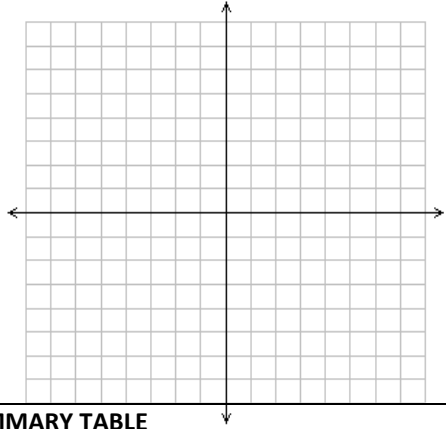
3.2 THE RECIPROCAL OF A QUADRATIC FUNCTION

- Reciprocal functions** can be analyzed using the key features: asymptotes, intercepts, domain, range, and positive and negative intervals.
- The behaviour(s) near the asymptote(s) is similar to that of reciprocal linear functions, and can be predicted by analyzing the **roots** of the quadratic relation in the denominator.

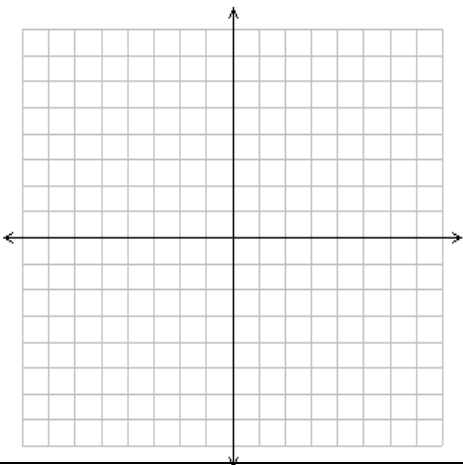
Case 1: RECIPROCALs of QUADRATIC FUNCTIONS with **no** x-intercept have **no** vertical asymptote. The graph is continuous and tends towards the horizontal asymptote.

$g(x) = \frac{1}{x^2 + 4}$	x-intercept	GRAPH 
	y-intercept	
	vertical asymptote	
	horizontal asymptote	
	Domain	
	Range	

Case 2: RECIPROCALs of QUADRATIC FUNCTIONS with **one** x-intercept have **one** vertical asymptote. The graph has 2 branches tending towards the vertical asymptote.

$h(x) = -\frac{1}{x^2 - 4x + 4}$	x-intercept	GRAPH 												
	y-intercept													
	vertical asymptote													
	horizontal asymptote	SUMMARY TABLE <table><tr><th>as $x \rightarrow$</th><th>$f(x) \rightarrow$</th></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	as $x \rightarrow$	$f(x) \rightarrow$										
	as $x \rightarrow$		$f(x) \rightarrow$											
Domain														
Range														

Case 3: RECIPROCALs of QUADRATIC FUNCTIONS with **two** zeros have **two** vertical asymptotes. The graph has 3 branches, with the middle one reaching a local **MAXIMUM** or **MINIMUM** point. This point is equidistant from the 2 vertical asymptotes.

$f(x) = \frac{1}{x^2 - 4x - 12}$	x-intercept	GRAPH 														
	y-intercept															
	vertical asymptote															
	horizontal asymptote	SUMMARY TABLE <table border="1"> <tr> <th>as x →</th> <th>f(x) →</th> </tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> </table>	as x →	f(x) →												
	as x →		f(x) →													
Domain																
Range																

Detailed description of the function (look at the graph and summary table):

⇒ reaches a LOCAL _____ on the interval _____ at x = _____ and y = _____.

⇒ f(x) increases when _____

⇒ f(x) decreases when _____

⇒ f(x) is positive when _____

⇒ f(x) is negative when _____

○ ○ ○
Note: "Sign of slope" is used when describing increasing & decreasing intervals in the branches of the

Try it now!

1. Suppose that n is a constant and that $f(x)$ is a linear or quadratic function defined when $x=n$. Complete the following sentences:

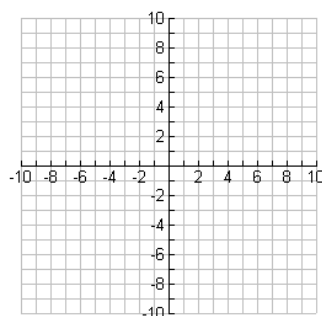
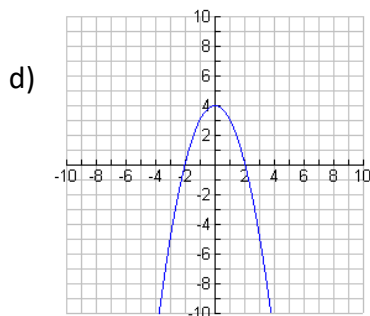
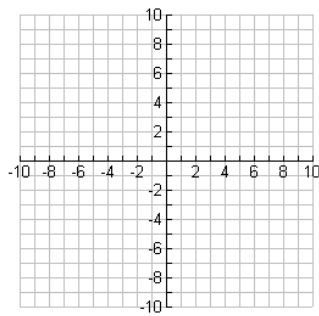
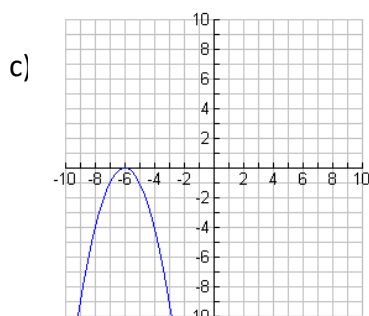
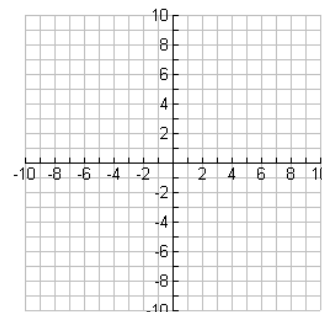
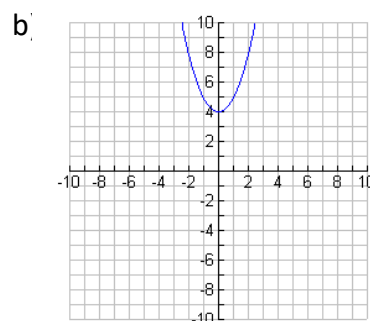
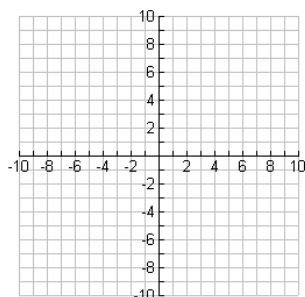
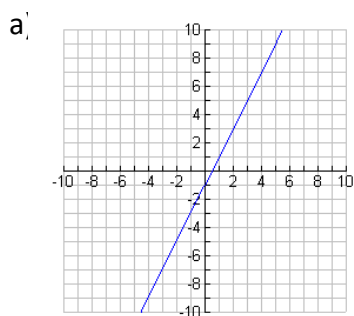
a) If $f(n)$ is large, then $\frac{1}{f(n)}$ is

b) If $f(n)$ is small, then $\frac{1}{f(n)}$ is ...

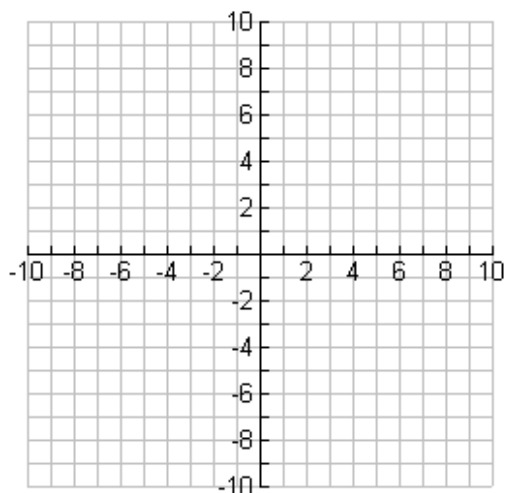
c) If $f(n)=0$, then $\frac{1}{f(n)}$ is ...

d) If $f(n)$ is positive, then $\frac{1}{f(n)}$ is ...

2. Using the graph of the function $y = f(x)$ provided, sketch the graph of the reciprocal function, $y = \frac{1}{f(x)}$.



3. Sketch the graph of the function, $f(x) = \frac{1}{x^2 + x}$.



i) domain _____

ii) range _____

iii) intervals of increase _____

iv) intervals of decrease _____

v) positive intervals _____

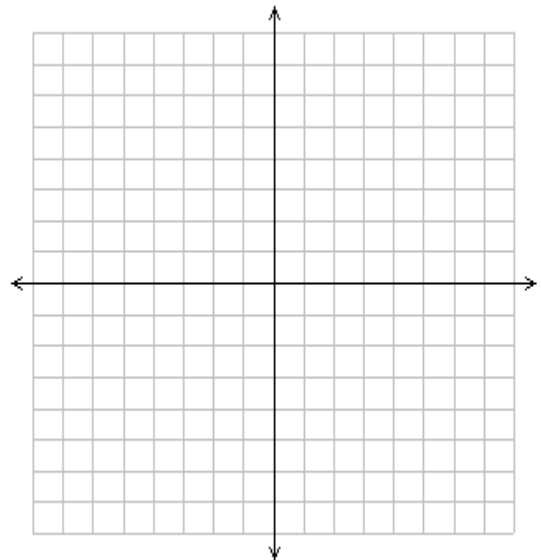
vi) negative intervals _____

3.3 RATIONAL FUNCTIONS OF THE FORM $f(x) = \frac{ax+b}{cx+d}$

KEY FEATURES of the rational function $f(x) = \frac{ax+b}{cx+d}$:

- The equation of the vertical asymptote can be found by setting the denominator **equal to zero** and solving for x, provided that the numerator does not have the same zero.
- The equation of the horizontal asymptote can be found by dividing each term in both the numerator and denominator by x, and by investigating the behaviour of the function as $x \rightarrow \pm\infty$.
- The constant **b** acts to stretch the curve, but it has **NO EFFECT** on the asymptotes, domain or range.
- The constant **d** shifts the vertical asymptote. The horizontal asymptote is also shifted.
- The 2 branches of the graph of the function are **equidistant** from the point of intersection of the vertical and horizontal asymptotes.

Ex 1: Given $f(x) = \frac{3x+2}{4x-1}$. Find the x- and y- intercepts. Find the equations of the vertical and horizontal asymptotes.
Sketch the function. State the domain and range.



SUMMARY of the rational function $f(x) = \frac{ax+b}{cx+d}$:

- | | |
|---------------------------------------|--|
| i) the y-intercept is $\frac{b}{d}$ | iii) the horizontal asymptote is $\frac{a}{c}$ |
| ii) the x-intercept is $-\frac{b}{a}$ | iv) the vertical asymptote is $-\frac{d}{c}$ |

Ex 2: Given $f(x)=\frac{x-2}{3x+4}$.

i) Determine the x- & y-intercepts, and the equations of the asymptotes.

ii) Without graphing, determine the positive and negative intervals of the function.

Hint: Let's examine the signs of the **numerator** and **denominator** and their **quotient**, to determine the pos^{ve}/neg^{ve} intervals.

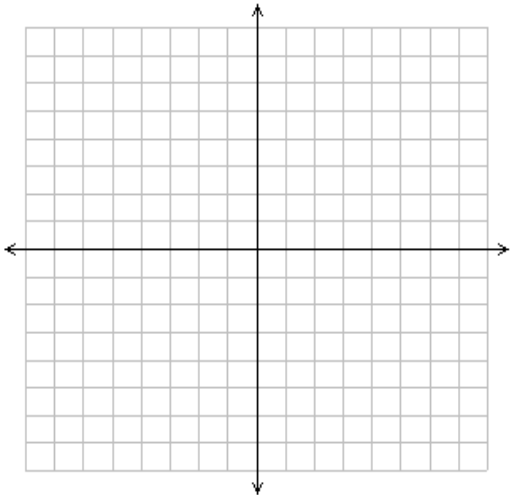
notes:

- ➡ the vertical asymptote and the x-intercept divide the x-axis into 3 intervals, what are they?
- ➡ **Intervals:** ● included (zeros)
 ○ not included (asymptotes)
- ➡ Choose #'s within each interval to evaluate the sign of each expression.

Numerator			
Denominator			
Quotient			

∴ f(x) is positive when:_____ and f(x) is negative when: _____.

- iii) Sketch f(x).
iv) State the domain and range.



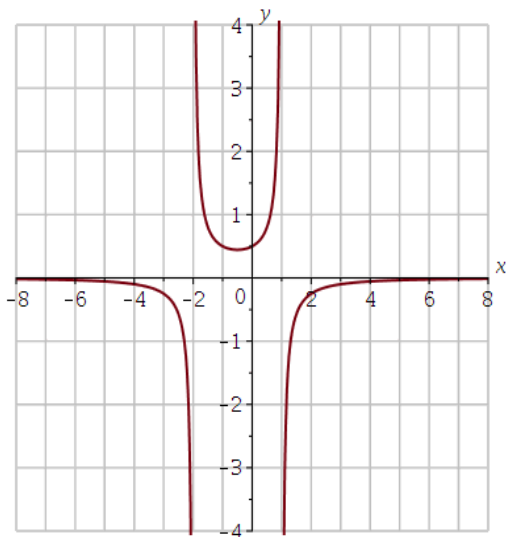
v) State the increasing and decreasing intervals of f(x)

Try it now!

Below is the graph of a reciprocal polynomial function, $y = \frac{1}{f(x)}$.

Which of the following is the equation of the function?

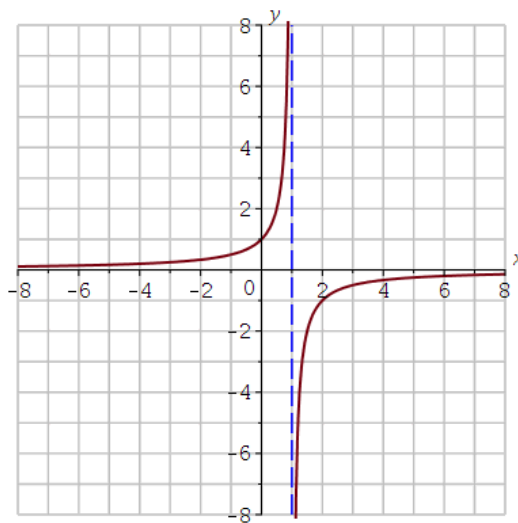
- ☐ $y = \frac{1}{(x+2)(x-1)}$
☐ $y = -\frac{1}{(x-2)(x+1)}$
☐ $y = \frac{1}{(x-2)(x+1)}$
☐ $y = -\frac{1}{(x+2)(x-1)}$



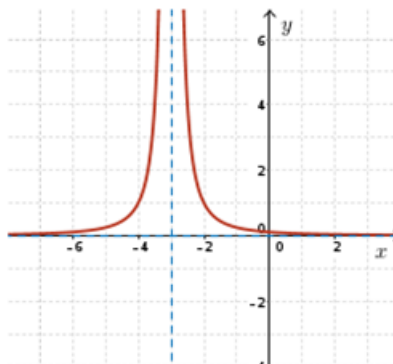
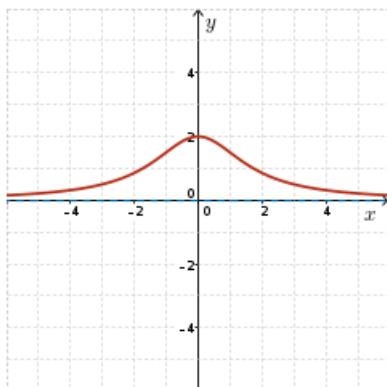
Below is the graph of a reciprocal polynomial function, $y = \frac{1}{f(x)}$.

Which of the following is the equation of the function?

- ☐ $y = \frac{1}{x-1}$
☐ $y = \frac{1}{x+1}$
☐ $y = \frac{1}{-x+1}$
☐ $y = \frac{1}{-x-1}$



3. State a possible equation for the two graphs below:



4.

Given the function $y = \frac{2x+1}{3x-1}$,

a) State the equation of the vertical asymptote: _____

b) State the equation of the horizontal asymptote: _____

c) Determine the x-intercept: _____

d) Determine the y-intercept: _____

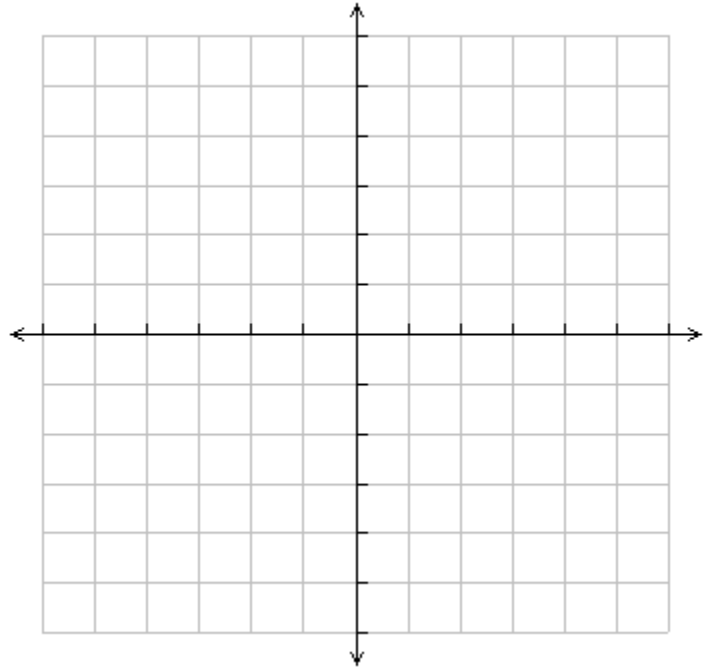
3.5 GRAPHING RATIONAL FUNCTIONS – SPECIAL CASES

CASE I: Simplify the equation by factoring to find a “typical” equation:

- Factor & reduce where possible.
- Indicate the restrictions on the variable in order to identify HIDDEN DISCONTINUITIES.
- Any restrictions in the denominator that have been factored out become HOLES.
- Any restrictions in the denominator that have not been factored out remain ASYMPTOTES.
- The factored numerator is used to determine all x-intercepts.

Example 1:

$$f(x) = \frac{x^2 + 8x + 16}{x^2 + x - 12}$$



Solution:

⇒ vertical asymptote:

⇒ horizontal asymptote:

⇒ x-intercept:

⇒ y-intercept:

⇒ discontinuity:

⇒ domain:

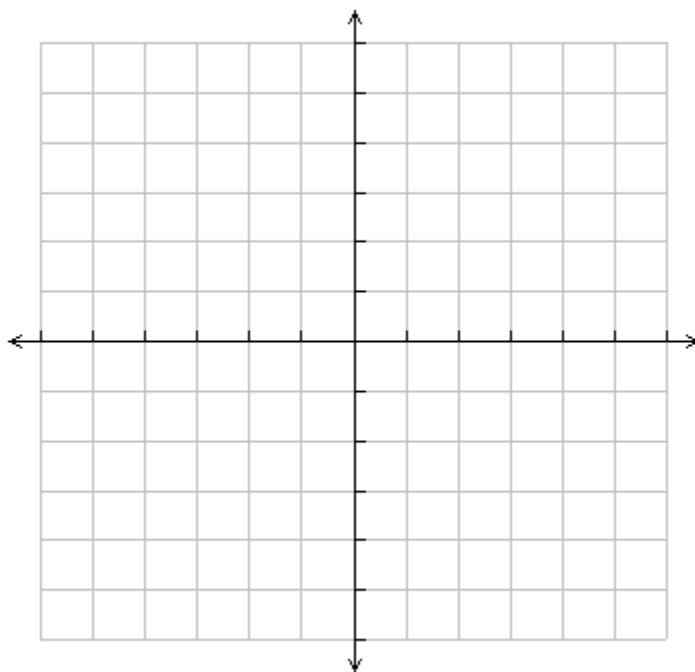
⇒ range:

⇒ increasing intervals:

⇒ decreasing intervals:

Example 2:

$$f(x) = \frac{x+3}{2x^3 - x^2 - 21x}$$



Solution:

⇒ vertical asymptote(s):

⇒ horizontal asymptote:

⇒ x-intercept:

⇒ y-intercept:

⇒ discontinuity:

⇒ local max/min:

⇒ domain:

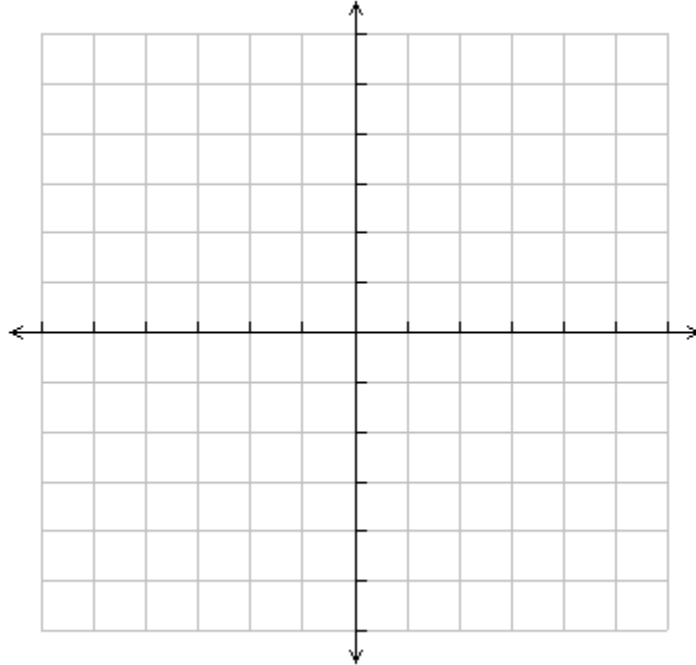
⇒ range:

CASE II: The rational equation cannot be simplified by factoring AND the numerator (degree 2) has higher degree than the denominator (degree 1):

- Factor the numerator to determine all x-intercepts.
- Any restrictions in the denominator remain vertical asymptotes.
- Use LONG DIVISION to determine the equation of the oblique or slant asymptote.

Example 3:

$$f(x) = \frac{x^2 - x - 12}{x - 3}$$



Solution:

⇒ vertical asymptote(s):

⇒ x-intercept(s):

⇒ y-intercept:

x	y

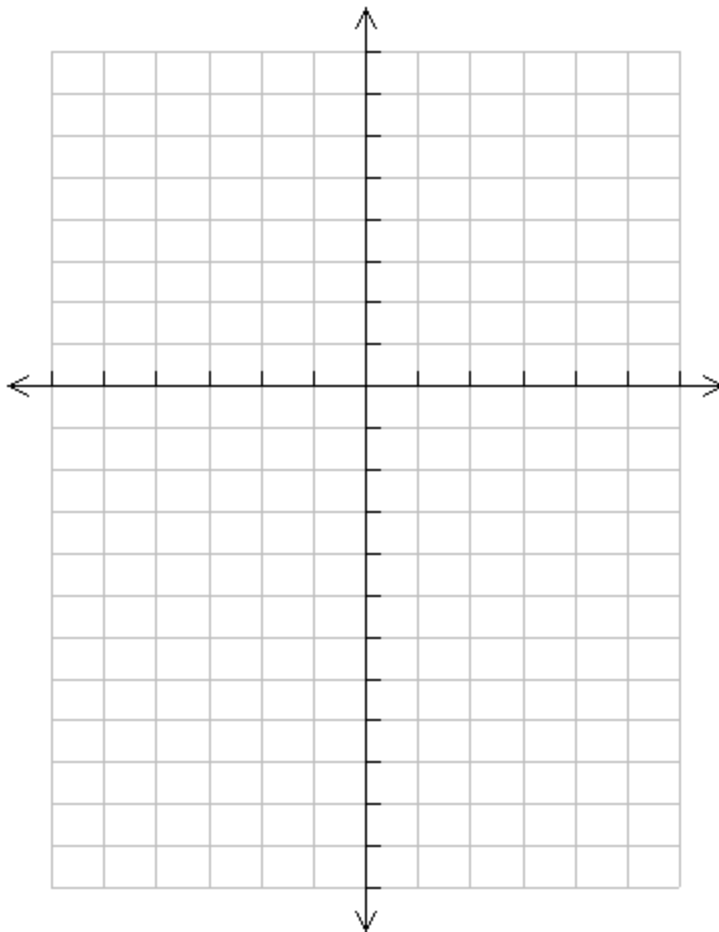
⇒ also has a “SLANT ASYMPTOTE” which is found by LONG DIVISION (or synthetic):

⇒ domain:

⇒ range:

Example 4:

$$f(x) = \frac{x^2 - 2x}{x + 1}$$



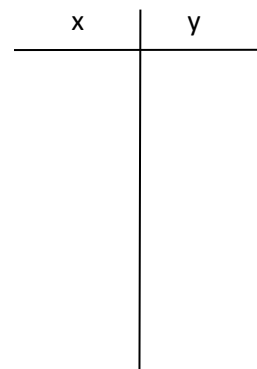
Solution:

⇒ vertical asymptote(s):

⇒ x-intercept(s):

⇒ y-intercept:

⇒ SLANT ASYMPTOTE:



⇒ domain:

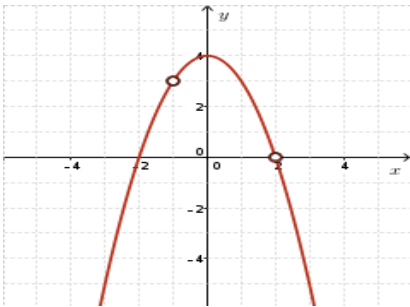
⇒ range:

Try it now!

1. Complete the comparison table below. Use this information, along with additional points, to predict the shape of the graph of each function. Verify your prediction using technology.

Function	$f(x)=\frac{x+1}{x^2-2x-3}$
Domain	
Vertical Asymptote(s) and/or Points of Discontinuity	
Horizontal Asymptote(s)	
x-intercepts	
y-intercepts	
Intervals where $f(x) > 0$	

2. Determine an equation for the rational function shown in the graph below.



3. The function $f(x)=\frac{2x^2+ax+b}{5x^2-26x+b}$, where a and b are real numbers, has a point of discontinuity (hole) when $x=6$.

- a. Determine the values of a and b.
- b. Determine the location of the hole, the x- and y-intercepts, and the equations of the asymptotes of the function.

3.4 SOLVING RATIONAL EQUATIONS AND RATIONAL INEQUALITIES

⇒ **STEPS TO SOLVING RATIONAL EQUATIONS ALGEBRAICALLY:**

- 1) FACTOR expressions in numerator & denominator to find common factors & restrictions.
- 2) Find the LCD and combine multiple terms into one term only on either side of the equality.
- 3) Solve using a variety of techniques (i.e. cross-multiply, simplify, quadratic formula, etc).

Ex. 1: Solve: $\frac{x}{x-2} + \frac{1}{x+4} = \frac{2}{x^2 - 6x + 8}$

Ex. 2: Solve: $-\frac{1}{x} = x^2 - 2$

⇒ **STEPS TO SOLVING RATIONAL INEQUALITIES ALGEBRAICALLY:**

- Re-write the inequality as ONE quotient with the right side equal to 0.
- Use the numerator to determine all zeros.
- Use the denominator to determine all vertical asymptotes.
- Then, use test points to determine the sign of the expression within each interval.

Ex. 1: Solve: $\frac{x^2 - 3x - 4}{x^2 + 11x + 30} \geq 0$

Solution:

- FACTOR the numerator to find the x-intercepts: _____
- FACTOR the denominator to find the restrictions: _____
- The function can only change sign at the x-intercepts or at the vertical asymptotes.
So we use all of these values to determine the positive/negative intervals of the function.

Intervals:

- included (zeros)
- not included (asymptotes)

We want the intervals where ≥ 0

(Note: we can always check our solution with a graphing calculator for a visual confirmation.)

Ex.2: Solve: $\frac{2}{x+3} \leq \frac{1}{x-1}$

Or the solution can be written as:

3.4 Extra Practice: Solving Rational Equations/Inequalities

Solve for x.

a) $\frac{1}{20} + \frac{1}{40} = \frac{1}{x}$

b) $\frac{1}{27} + \frac{1}{17} = \frac{1}{x}$

c) $\frac{1}{2x} + \frac{1}{x} = \frac{1}{2}$

d) $\frac{1}{3} + \frac{1}{4} = \frac{1}{x}$

e) $\frac{1}{4} - \frac{1}{x} = \frac{1}{5}$

f) $\frac{2}{t} + 14 = \frac{3}{2t}$

g) $\frac{5}{x} - \frac{9}{3x^2} = \frac{16}{x}$

h) $\frac{x}{x+5} - \frac{x-5}{x+10} = 0$

i) $\frac{3}{x-2} + \frac{4}{x+5} = \frac{2x+5}{x^2+3x-10}$

j) $\frac{2}{2x} - \frac{1}{x} = \frac{1}{2x^2}$

k) $\frac{2x}{9x^2-16} - \frac{1}{3x+4} = 36$

l) $\frac{x^2-3x-4}{x^2+11x+30} \geq 0$

m) $\frac{x-8}{x} \leq 3-x$

n) $\frac{x^2-x-6}{x-3} > 1$

Answers: a) $\frac{40}{3}$ b) $\frac{459}{44}$ c) 3 d) $\frac{12}{7}$ e) 20 f) $-\frac{1}{28}$ g) $-\frac{3}{11}$ h) $-\frac{5}{2}$ i) $-\frac{2}{5}$

j) no solution k) ± 1.34 l) $x < -6, -5 < x \leq -1, x \geq 4$ m) $x \leq -2, 0 < x \leq 4$ n) $x > -1, x \neq 3$

Try it now!

Based on your understanding of vertical asymptotes and holes, choose the most appropriate graph for each equation below. Support your choice.

a. $f(x) = \frac{x-3}{x^2-3x}$

b. $g(x) = \frac{x^2+3x}{x^2+9}$

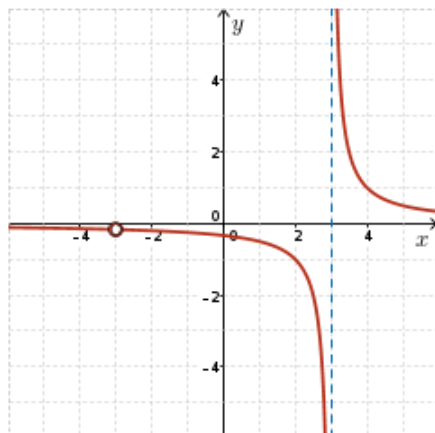
c. $h(x) = \frac{-3}{x+3}$

d. $p(x) = \frac{3x}{x^2+3x}$

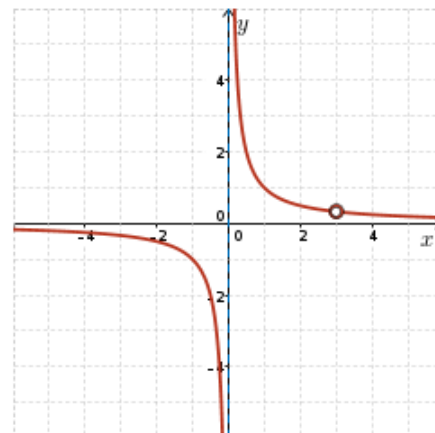
e. $q(x) = \frac{x+3}{x^2-9}$

f. $r(x) = \frac{3x+9}{x^2+6x+9}$

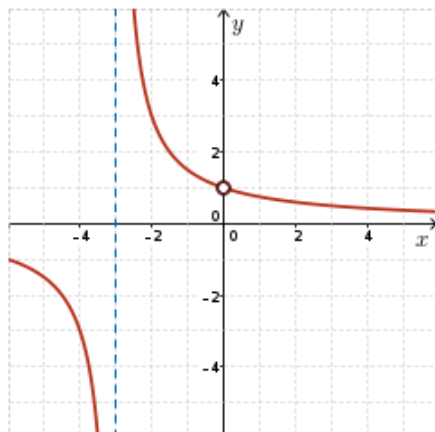
I.



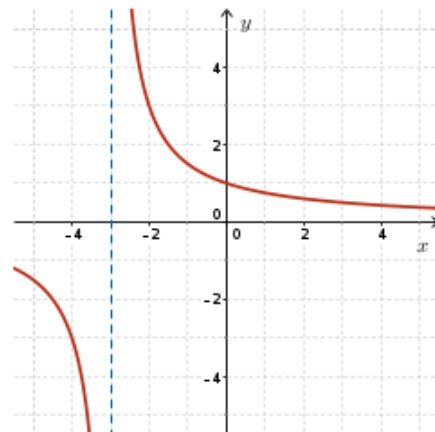
II.



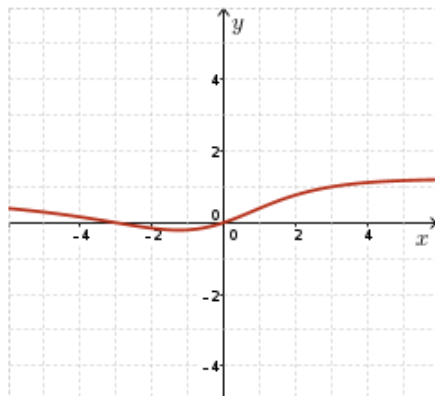
III.



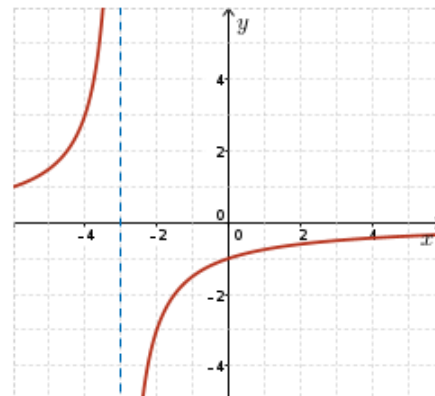
IV.



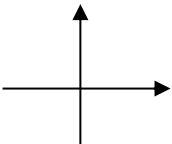
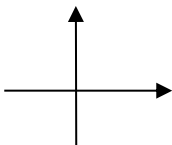
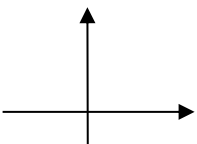
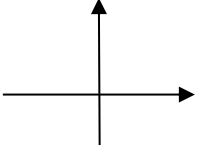
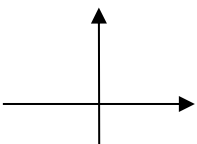
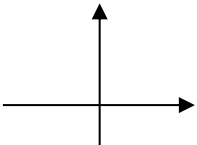
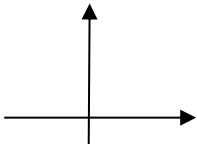
V.



VI.



Chapter 3 Summary Sheet: RATIONAL FUNCTIONS

Section	Type of Function	SKETCH	Characteristics
3.1	$f(x) = \frac{1}{kx - c}$		VA: from denominator: $x = \frac{c}{k}$ HA: $y=0$
3.2	Ex. $f(x) = \frac{1}{x^2 + 4}$		VA: none HA: $y = 0$
	Ex. $f(x) = \frac{1}{(x - 2)^2}$		VA: one VA from denominator HA: $y = 0$
	$f(x) = \frac{1}{ax^2 + bx + c}$		VA: two VA's from denominator (factor or use Quadratic Formula) HA: $y = 0$ Local maximum/minimum point
3.3	$f(x) = \frac{ax + b}{cx + d}$		Zero: from numerator: $x = -b/a$ VA: from denominator: $x = -d/c$ HA: $y = \frac{a}{c}$
3.5	Special Case I $f(x) = \frac{(2x + 1)(x - 4)}{(2x + 1)(x + 2)}$		Factor and reduce: equation turns into 3.1, 3.2 or 3.3 HOLE/DISCONTINUITY where reduced by factoring
	Special Case II $f(x) = \frac{(x - 4)(x + 1)}{(x + 2)}$		Factoring leads to no simplification Zero(s) from the numerator VA(s) from the denominator Use long division to find the equation of the SLANT/OBLIQUE asymptote
3.4	Equality: solve by rearranging into two fractions; then cross-multiply (don't forget restrictions) Inequality: solve by rearranging into one fraction $\leq$ or ≥ 0 ; then create a table with zeros and asymptotes		
ALL GRAPHS	domain & range, increasing & decreasing intervals, positive & negative intervals		

1. Which of the following has a horizontal asymptote at $y = 0$?

- A** $f(x) = \frac{1}{3-x}$ **C** $f(x) = -\frac{1}{x+2}$
B $f(x) = \frac{1}{17x+4}$ **D** all of the above

2. What is true about the function $f(x) = \frac{1}{3x+5}$ as $x \rightarrow -\frac{5}{3}^+$?

- A** $f(x) \rightarrow 0$ **C** $f(x) \rightarrow -\infty$
B $f(x) \rightarrow \infty$ **D** $f(x)$ is undefined

3. What is the x-intercept of $f(x) = \frac{1}{3x-4}$?

- A** $-\frac{1}{4}$ **C** $\frac{1}{4}$
B $\frac{4}{3}$ **D** There is no x-intercept.

4. Which function has a domain consisting of real numbers?

- A** $f(x) = \frac{1}{x-25}$ **C** $f(x) = \frac{1}{x^2-25}$
B $f(x) = \frac{1}{x+25}$ **D** $f(x) = \frac{1}{x^2+25}$

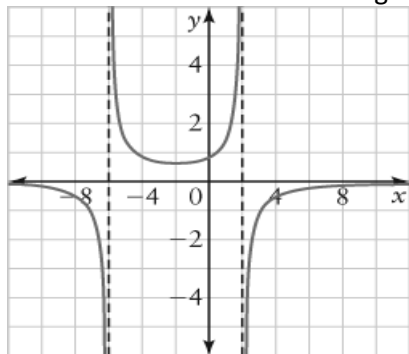
5. Which function has vertical asymptotes $x = -3$ and $x = \frac{1}{2}$?

- A** $f(x) = \frac{2}{4x^2+10x-6}$ **C** $f(x) = -\frac{1}{2x^2+5x-3}$
B $f(x) = \frac{1}{2x^2+5x-3}$ **D** all of the above

6. What is true about the function $f(x) = \frac{3}{x^2-6x+5}$, as $x \rightarrow -\infty$?

- A** $f(x) \rightarrow 0$ from below **C** $f(x) \rightarrow \infty$
B $f(x) \rightarrow 0$ from above **D** $f(x) \rightarrow -\infty$

7. Over what interval does the graph of the rational function have positive, decreasing slope?



- A** $x < -6$ **C** $-2 < x < 2$
B $-6 < x < -2$ **D** $x > 2$

8. Which function has a horizontal asymptote of $y = -\frac{2}{3}$?

- A** $f(x) = \frac{2x+3}{6x+1}$ **C** $f(x) = \frac{4x+3}{6x+1}$
B $f(x) = \frac{7x-2}{5x+3}$ **D** $f(x) = -\frac{4x+3}{6x+1}$